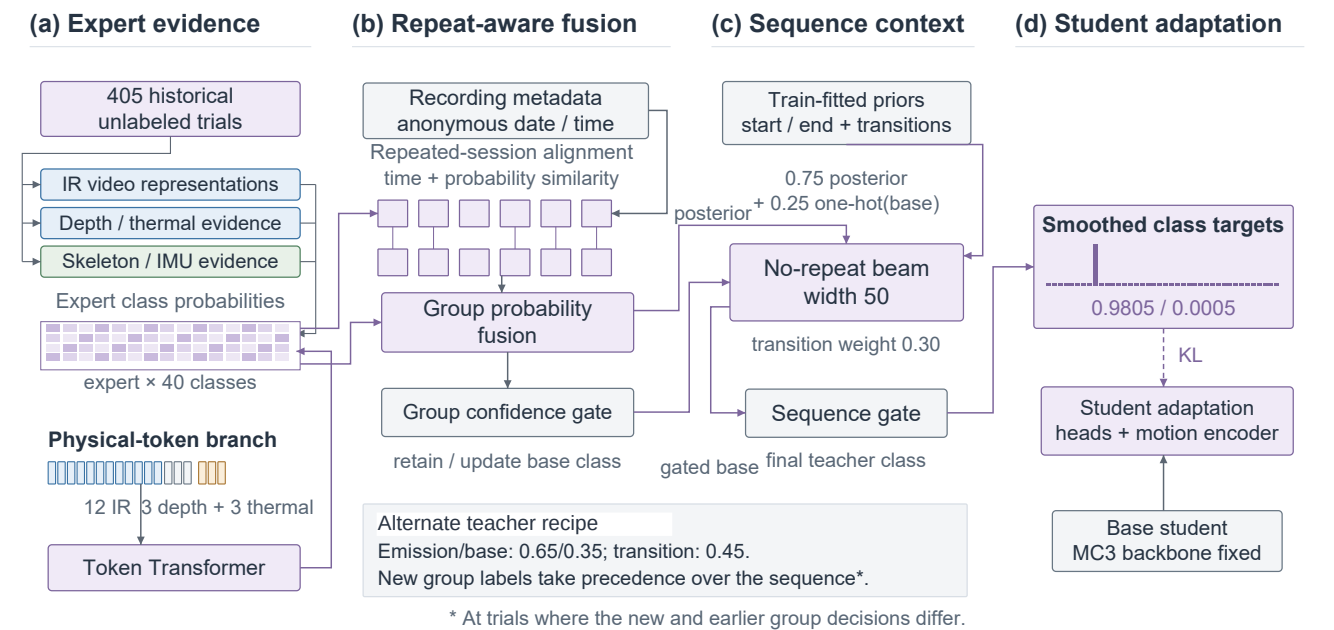

(a) Expert evidence
(b) Repeat-aware fusion
(c) Sequence context
(d) Student adaptation
405 historical unlabeled trials
IR video representations
Depth / thermal evidence
Skeleton / IMU evidence
Expert class probabilities
expert × 40 classes
Physical-token branch
12 IR
3 depth + 3 thermal
Token Transformer
Recording metadata anonymous date / time
Repeated-session alignment time + probability similarity
Group probability fusion
Group confidence gate
retain / update base class
Train-fitted priors start / end + transitions
0.75 posterior + 0.25 one-hot(base)
posterior
No-repeat beam width 50
transition weight 0.30
Sequence gate
gated base
final teacher class
Smoothed class targets
0.9805 / 0.0005
KL
Student adaptation heads + motion encoder
Base student MC3 backbone fixed
Alternate teacher recipe
Emission/base: 0.65/0.35; transition: 0.45.
New group labels take precedence over the sequence*.
* At trials where the new and earlier group decisions differ.